TEST REPORT

Overfill prevention sensor (PTC) — functional test

Report no. GWG-20260919-05

1 Unit under test

Type	GWG 4
Manufacturer	—
Serial no.	1
Year	—
Installation	Heizöltank 1

2 Test conditions

Date / Time	2026-09-19 14:12
Ambient temperature	21.0 °C
Relative humidity	60 %
Supply voltage	18.94 V
Series resistor	159,1 Ohm
Test mode	FIELD — probe left in the tank
Tested by	doctormord

Instruments and traceability

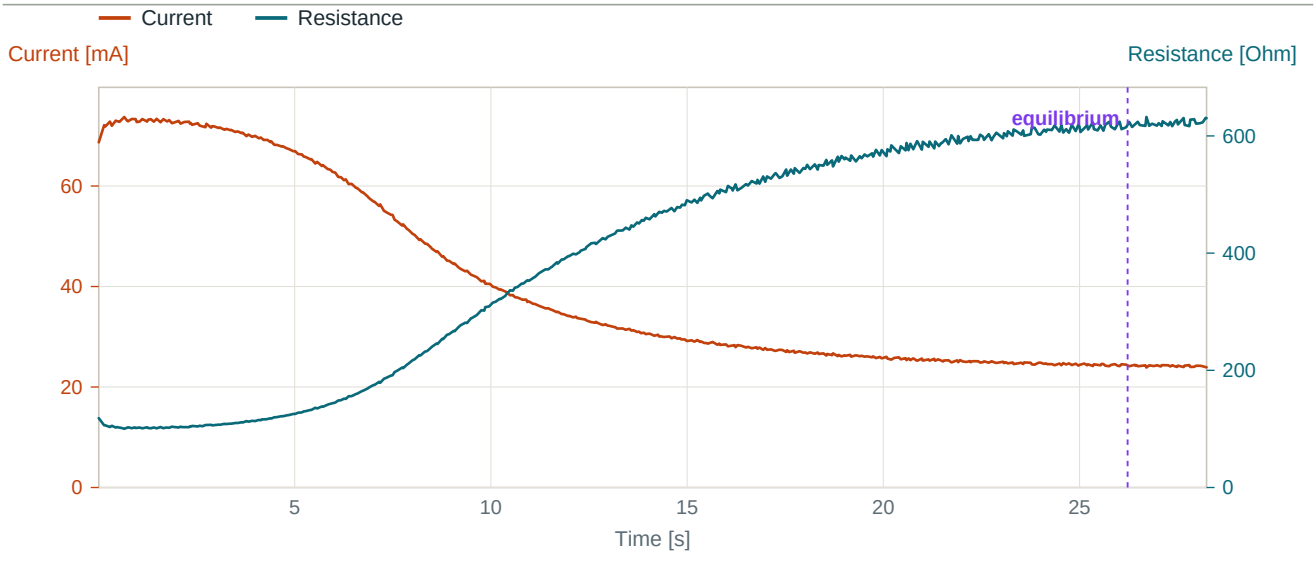
Spannungskette gegen Referenz-Multimeter kalibriert; Vorwiderstand durch direkte Strommessung bestimmt. Vor jedem Lauf wird der Kanal gegen die Versorgung genullt (Auto-Null). Restabweichung der Spannungskette ±0,04 V.

3 Result

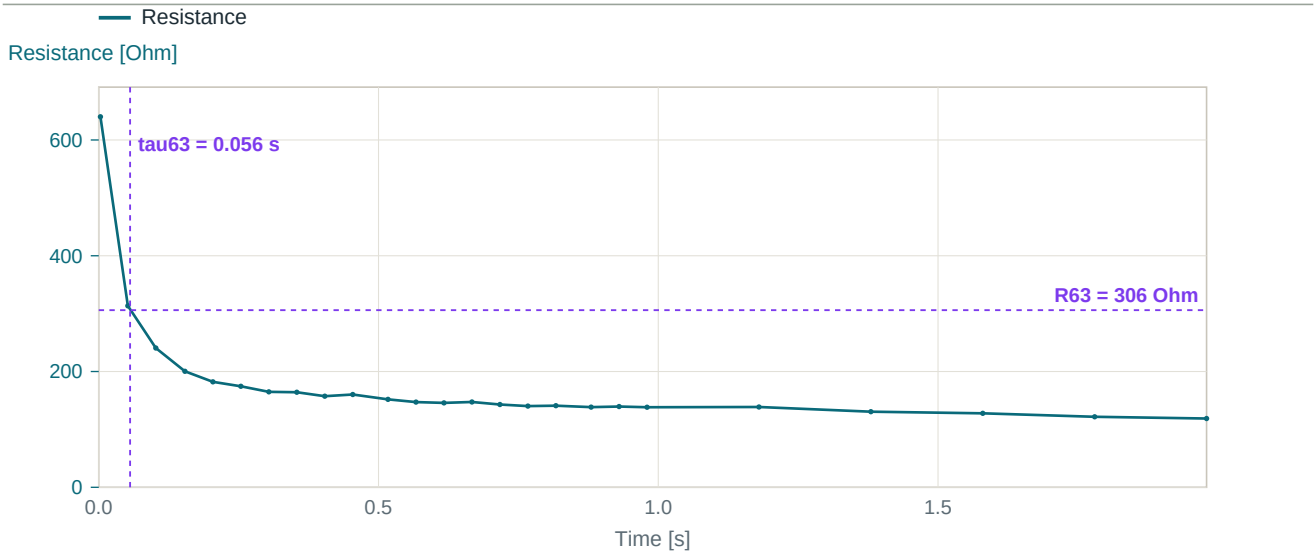
Quantity	Measured	Requirement	Verdict
Heat-up time to thermal equilibrium	26.2 s	max. 45 s	**pass**
Cold resistance (short pulse)	117.8 Ohm	—	
Hot resistance at equilibrium	630 Ohm	—	
Cold current	68.96 mA	—	
Hot current at equilibrium	24.14 mA	—	
Time constant tau63 in air	0.056 s	no standard value	**reference**
Time back to the cold value in air	1.98 s	no standard value	**reference**
Shutoff time after immersion	—	max. 1.5 s	**not tested**

OVERALL: PARTIAL — shutoff time not tested

4 Heating curve



5 Cooling curve in air



measuring pulses: 2.4 ms / 50 ms = 4.8 %

6 Method and measurement uncertainty

The PTC probe is connected to a DC supply through a series resistor and switched on the low side. Two voltages are measured: the supply and the node between series resistor and probe. Both channels are sampled interleaved so that supply ripple cancels in the difference. The current follows as I = (Vsup - U_A)/R_s - U_A/R_d, the probe resistance as R = U_A/I.

The heating phase does not end at a fixed current but at thermal equilibrium: the current counts as constant once its slope stays below 0.15 mA/s over three consecutive seconds. The heat-up time is taken as the start of the first quiet window, not the end of the third.

During air cooling the current is switched off and measured only in short pulses. The on-time is measured in the instrument with micros() and stated in this report; above 8 % duty the measuring pulses reheat the probe and the curve is no longer valid.

The air cooling curve has no standard value. It is a reproducible characteristic of the individual probe, intended for comparison against a known good probe of the same type.

The shutoff time after immersion can only be determined with the probe removed. It is not tested in FIELD mode.

7 Remarks

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8 Confirmation

The test was carried out using the method described in section 6. The values listed are those recorded directly by the test set.

Place, date

Signature